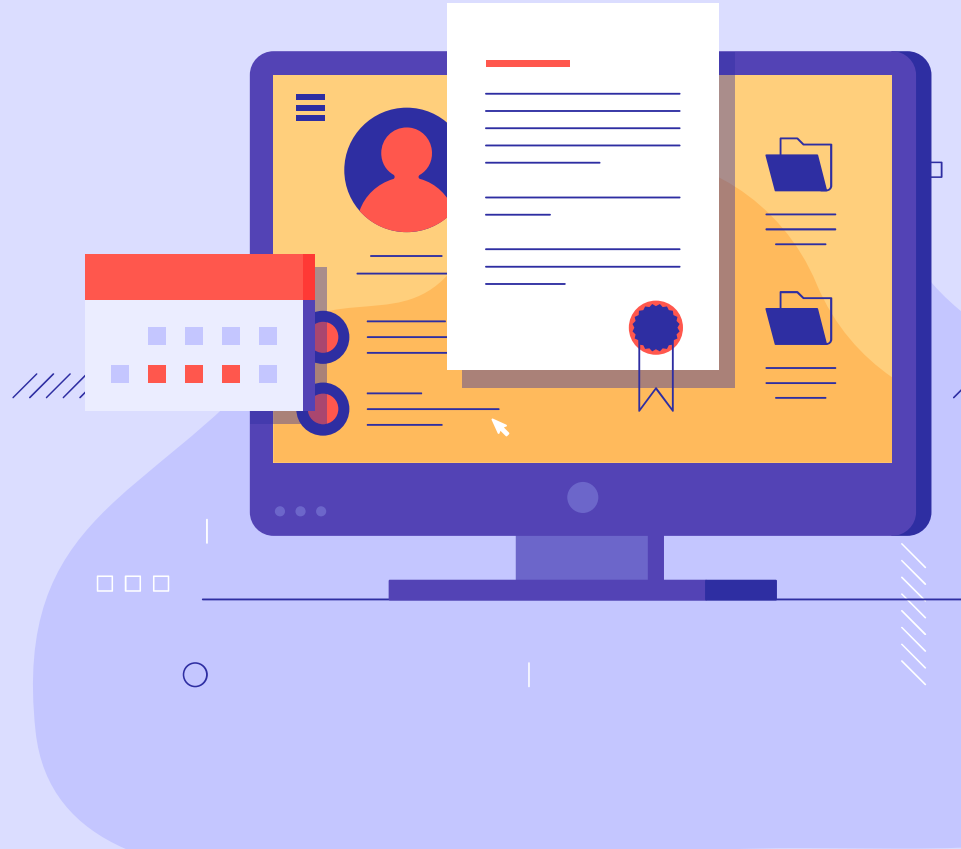


Resume/ Job Scanner

Carlos Colin-Feliciano

BIA 485 Applications of AI in Business - May 5th



Project Introduction

Many applicants apply to jobs without knowing whether their resume actually matches the role. Additionally, many companies are now using AI to quickly scan resumes and determine whether an applicant is a good fit for a position.

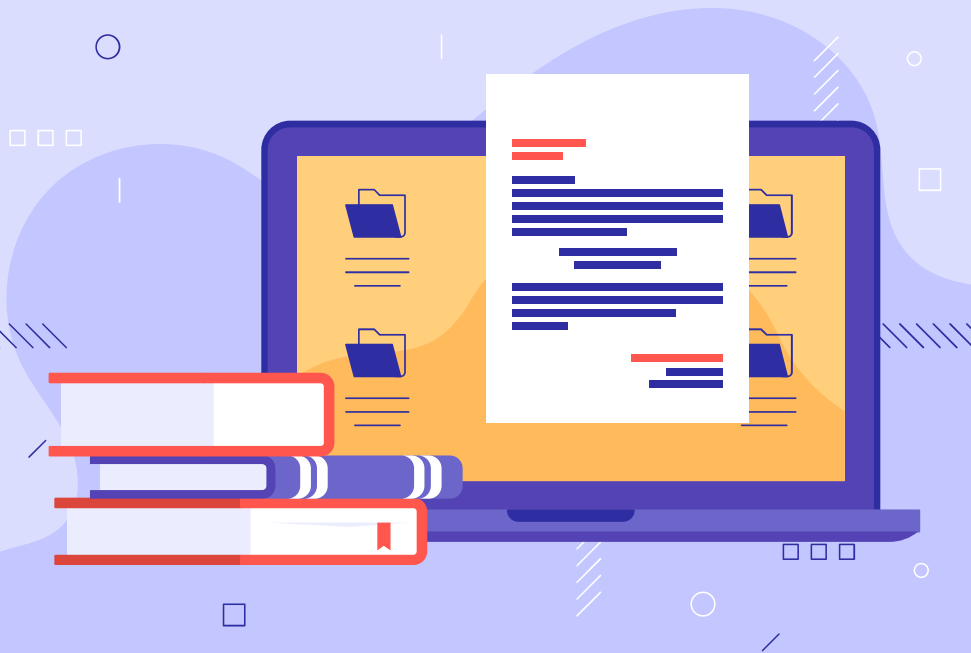


Problem Definition

Problem : Job applicants often struggle to know how well their resume matches a specific job description.

Motivation : Many companies use automated resume screening tools to quickly scan resumes for skills, keywords, and experience before a recruiter reviews them.

Real-World Application : This project uses NLP and machine learning to classify resume-job matches as No Fit, Potential Fit, or Good Fit, helping applicants improve their resumes before applying.



Dataset Description

SOURCE

HuggingFace open-source dataset
6.24k rows

TYPE

Text-based NLP classification dataset

MAIN COLUMNS

- resume_text
- job_description_text
- label

LABELS

- No Fit
- Potential Fit
- Good Fit

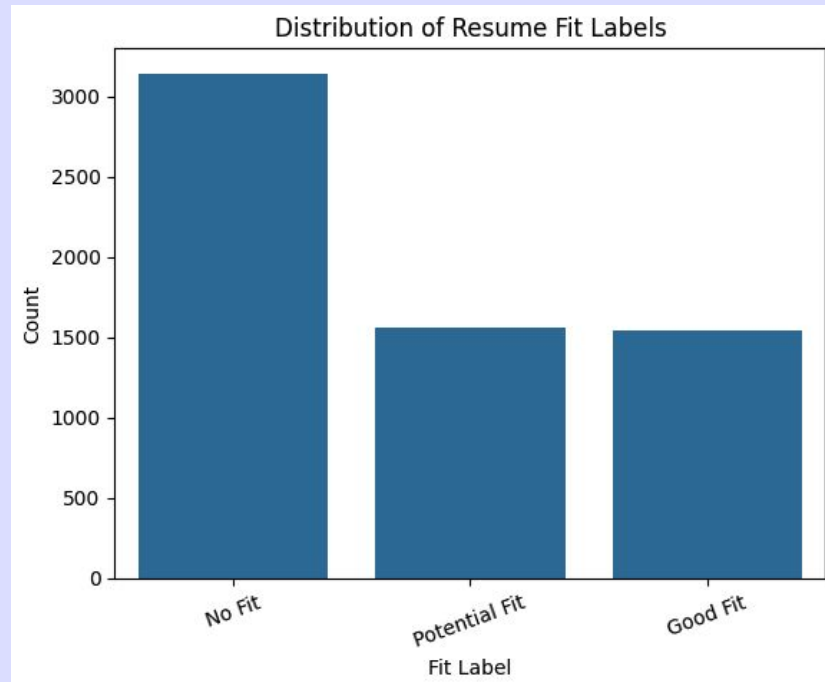
Exploratory Data Analysis

Key checks:

- Reviewed dataset shape, column names, and data types
- Checked for missing values in resume_text, job_description_text, and label
- Examined the distribution of the three fit categories
- Identified the target variable: label

The distribution of resume fit labels matters

- because it shows whether the dataset is balanced between “fit” and “not fit” examples. This helps ensure the model is not biased toward one label and that the accuracy results are meaningful.



Dataset Preprocessing

DATA CLEANING

Text data had to be cleaned before it could be used in a machine learning model.

Preprocessing steps:

- Converted all text to lowercase
- Removed punctuation, symbols, and special characters
- Removed extra spaces
- Created cleaned resume and job description columns
- Combined resume and job description into one feature
- Converted text into numerical features using TF-IDF

Raw Resume + Job Description



Text Cleaning



Combined Text Feature



TF-IDF Vectorization



Machine Learning Model

Pipeline Overview

01

DATA LOADING

Load resume and job description dataset with labeled fit categories (No Fit, Potential Fit, Good Fit).

02

DATA CLEANING

Clean text by converting to lowercase, removing punctuation, special characters, and extra spaces.

03

FEATURE ENGINEERING

Combine resume and job description text and convert into numerical features using TF-IDF.

04

MODEL TRAINING & EVALUATION

Train classification models and evaluate performance using accuracy, precision, recall, and F1-score.

Model Explanation: TF-IDF + Logistic Regression



TF-IDF

Converts resume and job description text into numerical features based on word importance.

- Weights important words higher
- Reduces importance of common words
- Helps capture key skills and keywords



LOGISTIC REGRESSION

Uses the features to classify each resume-job pair into one of the 3 label categories.

- Linear classification model
- Fast and efficient for text data
- Provides a strong baseline for prediction
- Works well with TF-IDF features

Model Comparison

Accuracy = Total Number of Predictions / Number of Correct Predictions



LOGISTIC REGRESSION: 63.6%

- Baseline model
- Simple, fast, and effective for text data



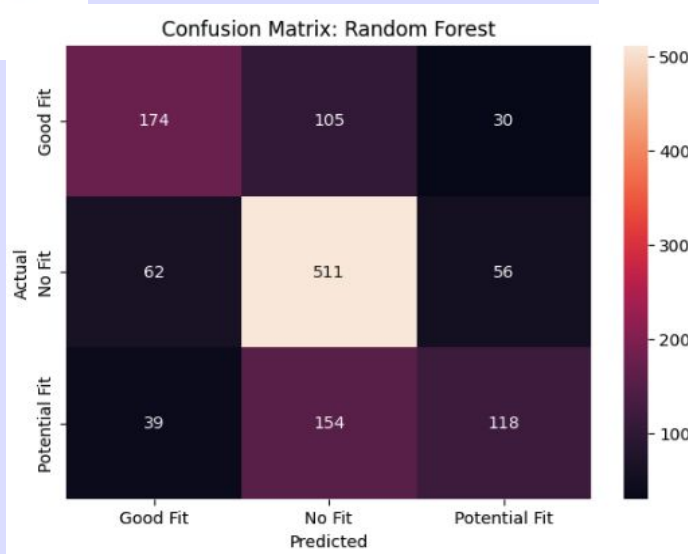
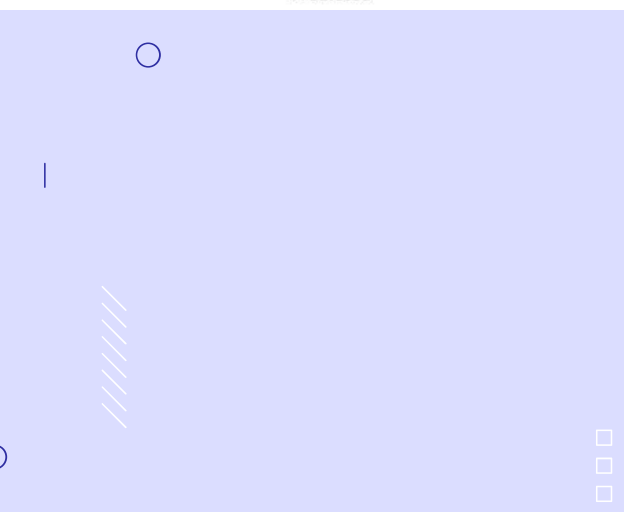
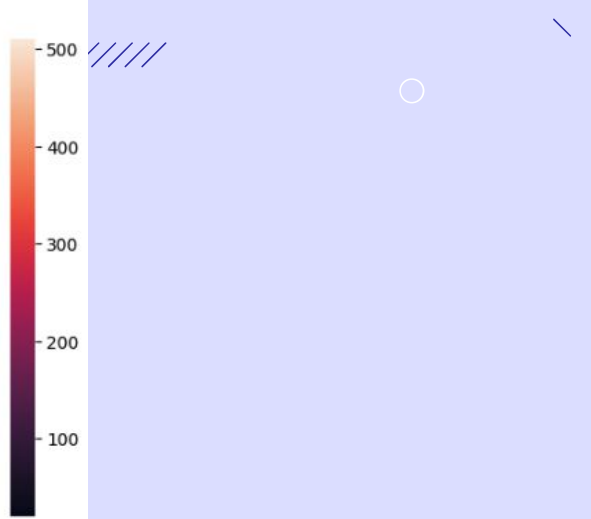
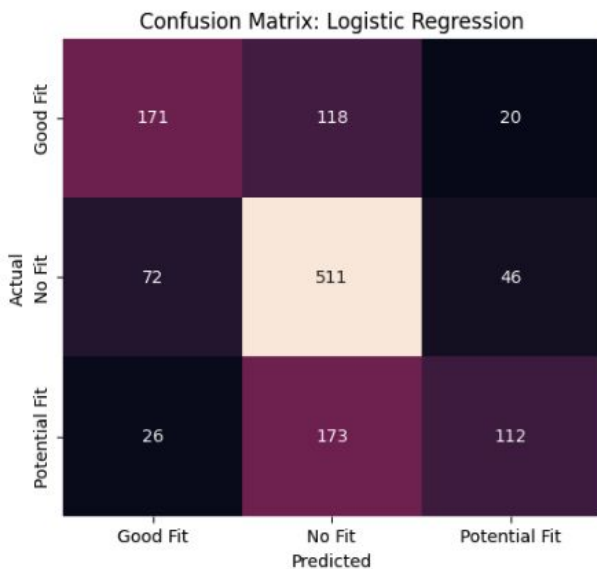
BALANCED LOGISTIC: 58.6%

- Adjusts for class imbalance
- Improves performance on weaker classes



RANDOM FOREST: 64.3%

- More complex, non-linear model
- Captures deeper patterns in data



Results / Model Improvements



BEST MODEL

Random Forest: 63.6%



KEY INSIGHTS

- Potential Fit Category is complex
- Strong No Fit identification



IMPROVEMENTS

- Tune Random Forest hyperparameters
- Improve text preprocessing
- Address class confusion with Potential Fit

Conclusion

This project shows that NLP can effectively analyze resume-job alignment, with Random Forest achieving the best performance at **64.3% accuracy**. However, classification was limited by overlap between categories, especially for “Potential Fit.”

FUTURE WORK & NEXT STEPS



Develop a user-friendly web application (HuggingFace)



Provide keyword-based resume improvement suggestions



Use larger and more diverse datasets



Explore advanced NLP models (RNN/LSTM & CNN)

Thank You!

Q&A

CREDITS: This presentation template was created by **Slidesgo**, including icons by **Flaticon**, infographics & images by **Freepik**

